

Estimating Gaussian Mixtures with the EM Algorithm

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1 EM Basics

Given observed data X and hidden variables Z and parameters θ defining a probabilistic graphical model $(\mathcal{G}, P)$. In order to estimate parameters that are most likely given the observed data we maximize the *log-likelihood function*

$$L(\theta) = \ln p(X|\theta) = \ln \sum_Z p(X, Z|\theta) \quad (1)$$

where we assume discrete Z . Since the logarithm is outside the sum over Z , $\max_{\theta} L(\theta)$ in general constitutes an intractable optimization problem. On the other hand, assuming we would know the latent variables Z , we could maximize the *complete-data* log-likelihood

$$L(\theta) = \ln p(X, Z|\theta). \quad (2)$$

Now the logarithm directly acts on the density, which for many distributions (e.g. the exponential family) yields tractable expression that can be maximized.

The standard Expectation Maximization (EM) algorithm ([Dempster et al., 1977](#)) makes use of this fact. Since Z is unknown and can not be used directly, it uses its expectation given X instead. Given an initialization θ_0 , EM consists of the following two steps:

$$\textbf{E-Step} : \quad \text{Evaluate } \mathbb{E}_Z[\ln p(X, Z|\theta)|X, \theta^{\text{old}}] := Q(\theta|\theta^{\text{old}}) \quad (3a)$$

$$\textbf{M-Step} : \quad \theta^{\text{new}} = \arg \max_{\theta} Q(\theta|\theta^{\text{old}}) \quad (3b)$$

Iteratively performing these steps is guaranteed to decrease the likelihood function (1) over the observed data X in each iteration. The algorithm terminates when $L(\theta)$ converges.

1.1 Maximize free variational energy

There is an alternative interpretation of EM, which also makes clear why the algorithm is guaranteed to increase $L(\theta)$ for each step unless the current θ

represents a local maximum. We can decompose the log-likelihood function for any function $q(Z)$ into

$$\ln p(X|\theta) = \mathcal{L}(q, \theta) + \text{KL}(q\|p), \quad (4)$$

where we have defined

$$\mathcal{L}(q, \theta) = \sum_Z q(Z) \ln \left\{ \frac{p(X, Z|\theta)}{q(Z)} \right\}, \quad (5a)$$

$$\text{KL}(q\|p) = - \sum_Z q(Z) \ln \left\{ \frac{p(Z|X, \theta)}{q(Z)} \right\}. \quad (5b)$$

From $\text{KL}(q\|p) \geq 0$ it follows that $\mathcal{L}(q, \theta)$ is a lower bound on $\ln p(X|\theta)$. It holds that

$$\mathcal{L}(q, \theta) = \ln p(X|\theta) - \text{KL}(q\|p) \leq \ln p(X|\theta). \quad (6)$$

This lower bound is also known as the *free variational energy*.

Both steps of the EM algorithm can be interpreted as maximizations of the lower bound $\mathcal{L}$. The **E-step** constitutes the maximization of $\mathcal{L}$ with respect to q while holding θ fixed at θ^{old} . Using the representation in (6), we see that since $\ln p(X|\theta)$ does not depend on q , the optimal solution that minimizes $\text{KL}(q\|p)$ is to set $q(Z)$ equal to $p(Z|X, \theta^{\text{old}})$, which makes the bound tight. Furthermore, this step does not change $L(\theta)$.

In the **M-step** we hold $q(Z)$ fixed and maximize $\mathcal{L}$ with respect to θ . Looking at the representation given in (5a), this corresponds to maximize the expectation of the complete-data log-likelihood with respect to $q(Z)$. Setting $q(Z) = \ln p(X|Z, \theta^{\text{old}})$ we obtain $Q(\theta|\theta^{\text{old}}) + H[q(Z)]$, c.f. (3a). This step will increase $\mathcal{L}(q, \theta)$ and, since $\text{KL}(q\|p)$ is zero, also $\ln p(X|\theta)$. However, since we hold $q(Z)$ fixed, the Kullback-Leibler divergence after that step will be $\text{KL}(p(Z|X, \theta^{\text{old}})\|p(Z|X, \theta^{\text{new}}))$, which will be non-zero.

The **M-step** constitutes a convex optimization, if the complete-data log-likelihood function is convex, which is the case if it belongs to the exponential family. Since $-\text{KL}(q\|p)$ is at its global maximum during that step, its gradient with respect to θ is zero, therefore the gradients of $\mathcal{L}(q, \theta)$ and $\ln p(X|\theta)$ at θ^{old} with respect to θ coincide.

2 Gaussian Mixture

The Gaussian mixture is given by

$$p(x) = \sum_k \pi_k \mathcal{N}(x|\mu_k, \Sigma_k) \quad \text{marginal} \quad (7a)$$

$$p(z) = \prod_k \pi_k^{z_k} \quad \text{prior} \quad (7b)$$

$$p(x|z) = \prod_k \mathcal{N}(x|\mu_k, \Sigma_k)^{z_k} \quad \text{class conditional} \quad (7c)$$

$$p(x, z) = \prod_k \pi_k^{z_k} \mathcal{N}(x|\mu_k, \Sigma_k)^{z_k} \quad \text{joint} \quad (7d)$$

where $z = \{z_1, \dots, z_K\}$ is an indicator vector determining the class affiliation of the point x .

Given a data set $X = \{x_1, \dots, x_N\}$ of i.i.d. data points, the log-likelihood function of a Gaussian mixture is

$$L(\pi, \mu, \Sigma) = \ln p(X|\pi, \mu, \Sigma) = \sum_n \ln \sum_k \pi_k \mathcal{N}(x_n|\mu_k, \Sigma_k), \quad (8)$$

whereas the complete data likelihood is

$$\ln p(X, Z|\pi, \mu, \Sigma) = \sum_n \sum_k z_{nk} \{\ln \pi_k + \ln \mathcal{N}(x_n|\mu_k, \Sigma_k)\}. \quad (9)$$

¹In the **E-step** we evaluate $\mathbb{E}_Z[\ln p(X, Z|\theta)|X, \theta^{\text{old}}] := Q(\theta|\theta^{\text{old}})$ using the current parameter estimates θ^{old} . This yields terms $\mathbb{E}_Z[z_{nk}|X, \theta^{\text{old}}]$. The posterior $p(Z|X, \theta)$ factorizes over n , such that

$$\begin{aligned} \mathbb{E}_Z[z_{nk}|X, \theta] &= \sum_Z z_{nk} p(Z|X, \theta) \\ &= \sum_{z_{nk} \in \{0,1\}} z_{nk} p(z_{nk}|x_n, \theta) \\ &= 1 \cdot \frac{p(z_{nk} = 1, x_n|\theta)}{\sum_{z_n} p(z_n, x_n|\theta)} \\ &= \frac{\pi_k \mathcal{N}(x_n|\mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_n|\mu_j, \Sigma_j)} := \gamma(z_{nk}). \end{aligned}$$

In the **M-step** we derivate $Q(\theta|\theta^{\text{old}})$ with respect to π_k , μ_k and Σ_k . For the derivative with respect to π_k we have to introduce an additional Lagrange term

¹In the subsequent elaboration, we will sometimes denote the set of parameters (π, μ, Σ) by θ for convenience.

to make sure that $\sum_k \pi_k = 1$. This yields a new set of parameters estimates:

$$\mu_k = \frac{1}{N_k} \sum_n \gamma(z_{nk}) x_n, \quad (10a)$$

$$\Sigma_k = \frac{1}{N_k} \sum_n \gamma(z_{nk}) (x_n - \mu_k)(x_n - \mu_k)^T, \quad (10b)$$

$$\pi_k = \frac{N_k}{N}, \quad (10c)$$

where $N_k = \sum_n \gamma(z_{nk})$. To initialize the set of parameters $\{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$, one usually uses the k-means algorithm.

2.1 Missing data

The EM approach naturally is able to handle randomly missing data by treating them as latent variables (Ghahramani and Jordan, 1994). For each data point x_n a subset of components may be missing. We denote that subset by x_n^m and the observed components by x_n^o . Using that notation, the complete-data likelihood is

$$\ln p(X^o, X^m, Z|\theta) = \sum_n \ln p(x_n^o, x_n^m, z_n|\theta). \quad (11)$$

The latter joint density can be split into the conditional Gaussian $p(x_n^o, x_n^m|z_n, \theta)$ and the prior $p(z_n|\theta)$. The expectation of the complete log-likelihood is

$$\begin{aligned} \mathbb{E}_{Z, X^m} [\ln p(X^o, X^m, Z|\theta) | X^o, \theta^{\text{old}}] \\ = \sum_n \sum_k \mathbb{E}[z_{nk}] \left\{ \ln \pi_k + \ln \mathcal{N} \left(\begin{bmatrix} x_n^o \\ x_n^m \end{bmatrix} \middle| \begin{bmatrix} \mu_k^o \\ \mu_k^m \end{bmatrix}, \begin{bmatrix} \Sigma_k^{oo} & \Sigma_k^{om} \\ \Sigma_k^{mo} & \Sigma_k^{mm} \end{bmatrix} \right) \right\}. \end{aligned} \quad (12)$$

Since now X^m is treated as a latent variable, the expectation also affects parts of the Gaussian related to x_n^m . Expressions for the following expectations have to be found: $\mathbb{E}[z_{nk}]$, $\mathbb{E}[z_{nk} x_n^m]$ and $\mathbb{E}[z_{nk} x_n^m x_n^{mT}]$.

The terms $\mathbb{E}[z_{nk}]$ are again $\gamma(z_{nk})$ but conditioned only on x_n^o :

$$\mathbb{E}[z_{nk}] = \frac{\pi_k \mathcal{N}(x_n^o | \mu_k^o, \Sigma_k^o)}{\sum_j \pi_j \mathcal{N}(x_n^o | \mu_j^o, \Sigma_j^o)} := \gamma^o(z_{nk}).$$

Let us denote the inverse covariance matrix by Λ . Then the next terms is:

$$\begin{aligned} \mathbb{E}[z_{nk} x_n^m] &= \sum_{z_{nk}} \int z_{nk} x_n^m p(x_n^m | x_n^o, z_{nk}, \theta_k) p(z_{nk} | x_n^o, \theta_k) dx_n^m \\ &= 1 \cdot p(z_{nk} = 1 | x_n^o, \theta_k) \int x_n^m p(x_n^m | x_n^o, z_{nk} = 1, \theta_k) dx_n^m \\ &= \mathbb{E}[z_{nk} | x_n^o, \theta_k] \cdot \mathbb{E}[x_n^m | x_n^o, \theta_k] \\ &= \gamma^o(z_{nk}) \cdot \{\mu_k^m - \Lambda_k^{mm^{-1}} \Lambda_k^{mo} (x_n^o - \mu_k^o)\}. \end{aligned}$$

Using similar reasoning, the expectation of the last term also factorizes:

$$\begin{aligned}\mathbb{E}[z_{nk}x_n^m x_n^{mT}] &= \mathbb{E}[z_{nk}|x_n^o, \theta_k] \cdot \mathbb{E}[x_n^m x_n^{mT}|x_n^o, \theta_k] \\ &= \gamma^o(z_{nk}) \cdot \{\text{cov}(x_n^m|x_n^o, \theta_k) + \mathbb{E}[x_n^m|x_n^o, \theta_k] \mathbb{E}[x_n^m|x_n^o, \theta_k]^T\}.\end{aligned}$$

We now can use these expectations as surrogates for X^m and Z when calculating new parameter estimates (10a)-(10c) in the **M-step**. Let $\tilde{x}_n$ be the vector x_n with x_n^m replaced by $\mathbb{E}[x_n^m|x_n^o]$. We then can write the modified update equations as

$$\mu_k = \frac{1}{N_k} \sum_n \gamma^o(z_{nk}) \tilde{x}_n, \quad (13a)$$

$$\Sigma_k = \frac{1}{N_k} \sum_n \gamma^o(z_{nk}) ((\tilde{x}_n - \mu_k)(\tilde{x}_n - \mu_k)^T + A_{nk}), \quad (13b)$$

$$\pi_k = \frac{1}{N} \sum_n \gamma^o(z_{nk}), \quad (13c)$$

where A_{nk} is the zero matrix except for the submatrix corresponding the missing components, which is $\text{cov}(x_n^m|x_n^o, \theta_k) = \Lambda_k^{mm^{-1}}$.

2.2 Experiments

This little experiment demonstrates the capability of EM to obtain good parameter estimates despite partly missing data. For this experiment, we randomly sampled $N = 100$ points from the 2-dimensional mixture distribution displayed in Fig. 1. Fig. 2 shows the estimates of EM for that data set. If we randomly set 30% of the entries in X to NaN, and re-estimate the parameters based in this new X , we obtain the estimates displayed in Fig. 3.

References

- Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the em algorithm. *J. R. Stat. Soc. Series B Stat. Methodol.*, 39(1):1–38.
- Ghahramani, Z. and Jordan, M. I. (1994). Supervised learning from incomplete data via an em approach. In *Advances in neural information processing systems* 6. Citeseer.

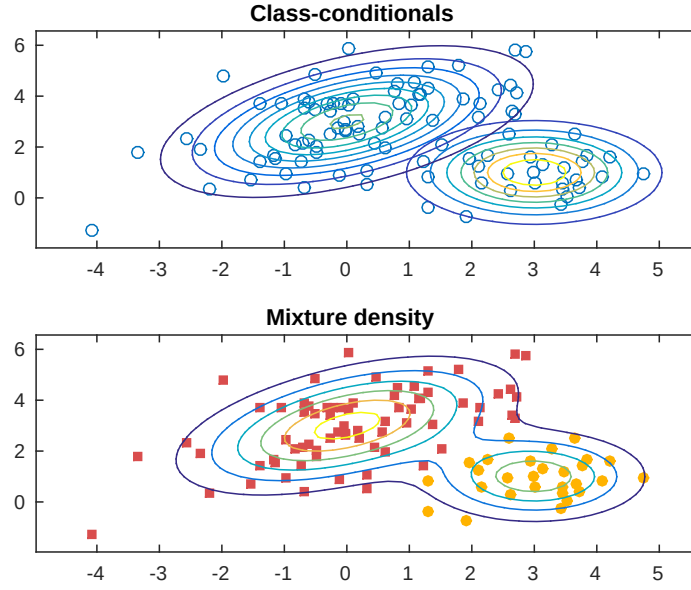


Figure 1: Underlying mixture density used to create the data set of $N = 100$ samples.

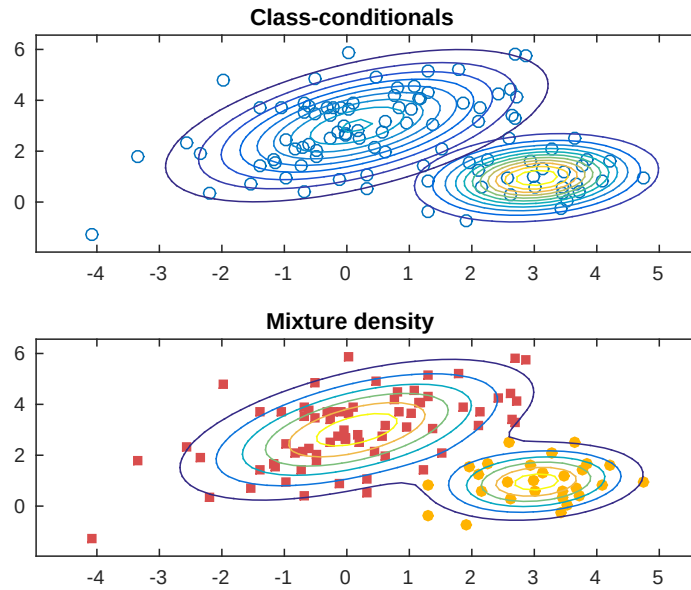


Figure 2: Estimated mixture based in X .

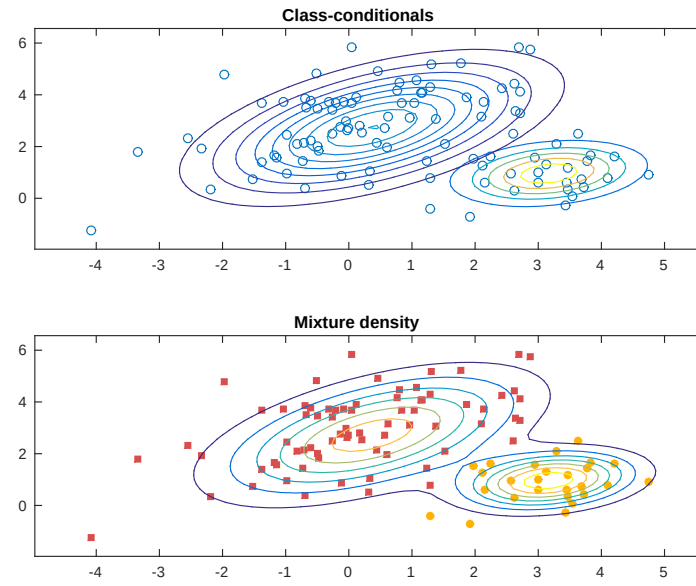


Figure 3: Estimated mixture based in X but this time with 30% of components in X set to NaN. The estimate is very similar to the one in Fig. 2 without missing data.